

RENSTON JAKE FERNANDES, Ph.D

Postdoctoral Researcher | Life Cycle Assessment | Techno-Economic Analysis | Process Modelling | Circular Economy | Emerging Low-Carbon Technologies

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PROFESSIONAL PROFILE

Ph.D. in Environmental Science and Engineering from KAUST with a strong process and chemical engineering profile focused on steady-state process simulation, carbon dioxide capture/utilisation pathways, life cycle assessment, techno-economic assessment, and decision-support analysis for low-carbon technologies. Experienced in developing ISO-aligned LCA and TEA workflows across production, operation, scale-up, and end-of-life considerations, with strong exposure to energy systems, circular economy, waste valorization, manufacturing processes, and commercialization-oriented feasibility assessment. Skilled in translating process data into environmental and economic indicators including GWP, acidification, eutrophication, cumulative energy demand, circularity, cost drivers, and scenario-dependent performance. Interested in applying model-based engineering and sustainability assessment to carbon dioxide removal and industrial decarbonisation technologies.

Core Competencies: Process modelling and simulation • Mass and energy balances • Techno-economic assessment • Sustainability and lifecycle assessment • Scale-up-oriented engineering analysis • Scenario and uncertainty analysis • Cross-functional collaboration • Technical communication

Current Toolkit: Environmental LCA • Foreground LCI development • Techno-economic analysis • Scenario analysis • Sensitivity and Monte Carlo uncertainty • Aspen Plus • GREET • GaBi (Sphera) • Python pipelines for reproducible sustainability assessment

EDUCATION

King Abdullah University of Science and Technology, Doctor of Philosophy (Ph.D) | *Environmental Science and Engineering*

11/2021 – 04/2026 | Thuwal, Saudi Arabia

Dissertation Topic: Circular Carbon E-Fuels from Biomass: Techno-Economic, Environmental, and Combustion Assessment Towards a Nation-Centric Sustainable Fuel Strategy.

- Developed integrated process and foreground inventory models for biomass-to-ethanol fermentation and e-methanol production pathways using Aspen Plus.
- Assessed circular conversion systems across region-specific feedstocks involving ethanol fermentation, biomass residue valorisation, CO₂ capture, and hydrogen-assisted methanol synthesis.
- Applied prospective, scenario-based sustainability assessment methods to emerging low-TRL production pathways, integrating environmental, economic, and circularity considerations for early-stage decision support.
- Conducted environmental life cycle assessment (LCA) using GREET, GaBi (Sphera), GLAM, and ReCiPe 2016 to assess the environmental burdens across the value chain, in accordance to ISO 14040/14044.
- Built Python-based workflows for scenario analysis, sensitivity analysis, and Monte Carlo uncertainty propagation.
- Performed techno-economic assessment of integrated production systems to support sustainability, scale-up, and commercial feasibility evaluation.
- Published and presented research at leading international conferences including CIRP LCE, EUBCE, SAE, and RWTH Aachen.

University of Sussex, Master of Science (M.Sc) | *Engineering Business Management*

09/2018 – 01/2020 | Brighton, United Kingdom

- Completed interdisciplinary coursework in strategic management, innovation and sustainability, marketing, and managing complex projects.
- Conducted applied research on project success evaluation for Palm Jumeirah Island and on the effects of corruption on foreign direct investment.
- Built strong foundations in economic evaluation, project management, and sustainability-oriented decision-making.

Master's Dissertation: Energy from Wastewater – An Economic Assessment of Wastewater in Skyscrapers.

Degree : Distinction (*top 1% of class; Sussex India Scholarship recipient*)

NMAM Institute of Technology,
Master of Technology (M.Tech) | Energy Systems Engineering

08/2014 – 06/2016 | Mangalore, India

- Graduated with Distinction, ranking in the top 1% of the class, demonstrating readiness for rigorous academic research.
- Completed a three-month internship at Lamina Suspension Products Ltd, gaining hands-on experience and conducting a Life Cycle Assessment (LCA) of a Leaf Spring Manufacturing Firm.
- Active member of the Solar Energy Society of India (SESI – Student Chapter), focusing on Renewable Energy & Alternative Energy Sources Combustion Engineering for sustainable energy solutions.

Master's Dissertation: Experimental Investigation of Cycle-by-Cycle (CBC) Fluctuations in a 4-Stroke SI Engine using Gasoline-Ethanol Blends.

Degree: Distinction

Sahyadri College of Engineering and Management,
B.E (Bachelor of Engineering) | Mechanical Engineering
Bachelor's Research Project : "Mechanically Operated Wheelchair cum Stretcher"

09/2010 – 06/2014 | Mangalore, India

Degree: First Class

SKILLS

Chemical Engineering and Process Engineering

Simulation Software – Expert

- Aspen Plus (process flow modeling, reactors, separation systems, recycle loops).
- Aspen Custom Modeler
- Aspen HYSYS
- Engineering Equation Solver (EES) – for cycle analysis and thermodynamic calculations.

Techno-Economic Analysis and Market Feasibility

Analysis of Emerging Energy Systems – Expert

- Capital cost estimation (PEC, CAPEX, OPEX using CEPCI indices)
- Discounted cash flow models (NPV, IRR, Payback Period, LCOM)
- Monte Carlo simulations for uncertainty and sensitivity analysis (Coding with Python) and excel spreadsheets.
- Scenario building and benchmarking with literature.
- Skilled in Python for techno-economic data analysis and risk assessment.

Scientific Research Methods and Communication

Outreach – Expert

- Excellent written and verbal communication skills, including writing reports, manuscripts, conference abstracts, and publication-ready academic material
- Excellent presentation skills, with experience to present data and research findings to academic, industrial, and wider professional audiences
- Strong literature review, research design, data collection, and data interpretation skills.
- Creative approach to problem-solving and development of new concepts

Teaching and Mentoring – Proficient

- Mentoring of students and provide guidance to interns by organizing training activities.
- Collaborative work in multidisciplinary teams.
- Workshop and training support.
- Project planning and proposal support.

Life Cycle Assessment – Expert

- LCA software: GREET, GaBi (Sphera) & Simapro
- Methodology: ReCiPe 2016, GLAM
- Integration of Random forest for pilot surrogate modelling for TEA/LCA outputs.
- Predictive modeling using NSGA-II for environmental impact analysis, process optimization and optimization strategies.

Programming Languages & Statistical Data

Analysis – Expert

- Python (data processing, Monte Carlo, pandas, NumPy, matplotlib, statistical analysis)
- Excel (advanced formulas, TEA model building).
- MATLAB for post-processing of Experimental data.

Interdisciplinary, Soft and Interpersonal Skills

– Expert

- Well-organized and an advocate of teamwork with a structured, analytical, and solution-oriented working style.
- Ability to work independently and cross-functionally in interdisciplinary, fast-paced, and dynamic team environments.
- Resilient, adaptable, and dependable contributor, able to maintain confidentiality and deliver effectively under evolving research demands.
- Cooperative and collaborative team player with emerging leadership skills, problem-solving skills and research delivery.
- Highly motivated professional with strong self-management, initiative, and a proactive approach to problem-solving.

PROFESSIONAL RESEARCH EXPERIENCE | WORK HISTORY

Srinivas Institute of Technology, Teaching Faculty -
Department of Marine and Mechanical Engineering

07/2016 – 05/2018 | Mangalore, India

Interdisciplinary Teaching & Curriculum Development

- Lecturing and tutoring modules including Fluid Mechanics and Non-Conventional Energy Sources, integrating practical and theoretical knowledge.
- Designed and delivered coursework on Energy & Environment, fostering a holistic understanding of sustainability.
- Developed curricula incorporating modern engineering tools such as Computer-Aided Machine & Marine Drawing, ensuring alignment with real-world engineering applications.
- Undertook undergrad student project supervision on waste plastic-to-fuel conversion and alternative fuel systems, contributing to early exposure to circular economy and materials-related sustainability topics.

Research Mentorship in Renewable Energy

- Undertook the task of supervising students at the undergraduate level in an innovative project on the "Experimental Investigation of Cycle-by-Cycle Fluctuations in a 4-Stroke SI Engine using Gasoline-Butanol Blends," focusing on performance optimization and thermal efficiency.
- Led two state-funded (KSCST) research projects: Waste plastic-to-fuel conversion, contributing to sustainability and circular economy efforts. Heavy Fuel Oil (HFO) blended with cooking oil, providing insights into alternative fuel viability.

Administrative & Regulatory Expertise

- Managed the procurement of project approvals and clearances from governmental agencies, ensuring compliance with regulatory standards.
- Conducted detailed analyses of departmental documents for accurate distribution and archival, improving institutional efficiency.
- Facilitated student-industry interactions for research funding and collaboration, emphasizing the practical applicability of academic research.

King Abdullah University of Science & Technology,
PhD Researcher

11/2021 – 04/2026 | Thuwal, Saudi Arabia

Affiliated with the Clean Energy Research Platform (CERP), formerly Clean Combustion Research Center (CCRC), under Prof. James W. G. Turner at the Advanced Vehicle Propulsion Laboratory (AVPL).

PhD Thesis Objective: Circular Carbon E-Fuels from Biomass: Techno-Economic, Environmental Sustainability, and Combustion Assessment Toward a Saudi-Centric Sustainable Fuel Strategy

- Conducted comprehensive well-to-tank LCA and TEA of synthetic e-fuels derived from regionally dominant agricultural biomass for conventional and novel engine applications.
- Developed integrated Aspen Plus models for biomass-to-ethanol fermentation and e-methanol production pathways, extracting performance indicators including conversion efficiency, energy intensity, CO₂ capture rate, and cost drivers across multiple regional feedstocks.
- Conducted well-to-tank life cycle assessment using GREET, GaBi (Sphera), GLAM, and ReCiPe methodologies, contributing to peer-reviewed publications.
- Performed multi-scenario techno-economic modelling of integrated biofuel and e-fuel plants to evaluate commercial feasibility, cost drivers, and sustainability trade-offs.
- Built reproducible Python workflows for scenario analysis, sensitivity analysis, KPI generation, and Monte Carlo uncertainty propagation.
- Experimentally tested hydrous ethanol-methanol blends in a CFR engine and applied structured experimental design to assess the effects of fuel blend composition and operating conditions on combustion and emissions performance using FTIR-based profiling.
- Evaluated integrated circular pathways involving ethanol fermentation, biomass residue valorisation, CO₂ capture, and hydrogen-assisted methanol synthesis.

Collaboration 1: Hydrogen-Fueled Variable Compression Ratio (VCR) HCCI Engine.

- Investigated and published two co-authored peer-reviewed studies on hydrogen as a zero-carbon fuel in a modified single-cylinder CFR engine under HCCI mode across compression ratios (14:1 to 18:1) and intake temperatures (110–150°C).
- Study 1: Investigated the effect of variable compression ratio (14:1–18:1) on hydrogen HCCI combustion in a modified CFR engine. Analyzed combustion phasing (CA₅₀), net IMEP, thermal efficiency (~33%), and emissions including near-zero NO_x under ultra-lean $\lambda = 4.8$ conditions.
- Study 2: Studied the influence of intake temperature (90–150°C) on hydrogen HCCI performance at fixed CR (17:1). Demonstrated expansion of lean limit to $\lambda = 5$, observed trade-offs in heat release, combustion duration, and knock sensitivity. Achieved thermal efficiency up to 37.9% and defined optimal operating conditions for future hydrogen engine design.

Collaboration 2: Life cycle assessment of ammonia and hydrogen as alternative fuels for marine internal combustion engines.

- Conducted a comprehensive life cycle assessment (LCA), from a well-to-wheel perspective for engine-integrated reformer systems.
- The fuel cycle was split into two phases including production of fuel or well-to-tank (WTT), blue ammonia via SMR and CCS, green hydrogen via electrolysis, feedstock transportation, and onboard storage.

Collaboration 3: Waste Heat Recovery from Marine and Truck Engines – A Joule Cycle-Based Approach.

- Conducted and submitted a comprehensive Techno-Economic Analysis comparing Open Joule Cycle and Organic Rankine Cycle (ORC) systems for energy recovery in heavy-duty vehicles.

★ INTERNSHIPS AND SUMMER PROGRAMS

Aramco (Saudi) - Summer Research Intern

06/2025 – 08/2025 | Ras Tanura, Saudi Arabia

Techno-Economic Analysis of a Novel Dual PSA-Based Flue Gas Separation System for E-Methanol Production

- Conducted a techno-economic evaluation of a patented dual-PSA flue gas separation and purification concept integrated with CO₂ hydrogenation to methanol using Aspen Plus.
- Developed a scenario matrix varying energy demand, conversion efficiency, process parameters and market drivers, and quantified uncertainty from hydrogen price, electricity cost, and catalyst life.
- Evaluated commercial feasibility, scenario-dependent business opportunities, and scale-up implications for industrial deployment.
- Prepared decision-support summaries for technical stakeholders, translating model outputs into concise evidence for applied industrial and non-academic use.

Radboud University Summer School - Summer Research Student

07/2024 – 07/2024 | Nijmegen, Netherlands

Advances in Environmental Life Cycle Modeling

- Gained advanced training in life cycle assessment, including inventory modelling, impact assessment, uncertainty analysis, and spatially differentiated environmental modelling.
- Worked with modelling approaches relevant to hydrology, ecology, pollutant impact assessment, and broader natural sciences-based environmental evaluation.
- Studied attributional and consequential LCA, allocation issues, and emerging approaches in circular economy and recycling assessment.
- Developed LCA models using MATLAB and Excel, applying LC-IMPACT and GLAM for spatially differentiated impact assessment.
- Built foundational exposure to impact category modelling relevant to toxicity-oriented and pollutant-related sustainability interpretation.

Lamina Suspension Products Ltd - Graduate Summer Intern

06/2015 – 09/2015 | Mangalore, India

Leaf Spring Energy Performance Analysis

Conducted an LCA-related assessment in a leaf spring manufacturing setting, gaining early industrial exposure to manufacturing systems, materials processing, heat-treatment operations, and process-level energy efficiency monitoring in a leaf spring production environment.

Process Calibration & Monitoring

- Conducted heat pumps calibration and oil flow monitoring in heating furnaces, showcasing attention to data accuracy and process monitoring, essential for Life Cycle Assessment (LCA) studies.

Performance Optimization

- Designed and implemented an employee evaluation process, leading to measurable performance improvements and showcasing skills in process efficiency and system optimization.

Zhejiang University - Research Trainee,

11/2025 – 11/2025 | Zhejiang, China

9th International Training Workshop on Waste-to-Energy

- The training covered advanced incineration and AI-enabled smart WtE, Waste-to-X system planning and optimization, MSW combustion fundamentals, and emissions control technologies (incl. dioxins and pollution control).
- Participated in sessions highlighting digital-twin-supported plant performance and operations and broader international perspectives on WtE policy and deployment.
- This training strengthened my ability to connect waste-derived resource recovery pathways with model-based decision support for early-stage system design and consistent feasibility assessment.

SELECTED RECENT RESEARCH OUTPUTS

Quantifying the potential of Saudi date palm fruit and its waste for e-methanol production using integrated process modelling, techno-economic analysis and life-cycle assessment., Elsevier 10/04/2026

Journal: Biomass and Bioenergy (Under Review)

Pre-print DOI: <https://dx.doi.org/10.2139/ssrn.6569602> 

Author(s): Renston Jake Fernandes, Mohammad Raghieb Shakeel, Ducduy Nguyen, Hong G. Im, James W. G. Turner

A Joule cycle-based approach to recover waste heat from marine and heavy-duty vehicles, Elsevier 04/10/2025

Journal: Energy

DOI: <https://doi.org/10.1016/j.energy.2025.138755> 

Author(s): Sreenivasa Rao Gubba, Raghu Vamsi Kodaboina, Kesty Y. Kenkoh, Renston Jake Fernandes, James W. G. Turner

Techno-Economic Analysis of an Integrated E-Methanol Production Plant Utilizing First and Second Generation Biomass Resources, SAE International 16/06/2025

Journal: SAE Technical Paper

DOI: <https://doi.org/10.4271/2025-01-0182> 

Author(s): Renston Jake Fernandes, Mohammad Raghieb Shakeel, Ducduy Nguyen, Hong G. Im, James W.G. Turner

Life cycle assessment of ammonia and hydrogen as alternative fuels for marine internal combustion engines, Elsevier 25/03/2025

Journal: International Journal of Hydrogen Energy

DOI: <https://doi.org/10.1016/j.ijhydene.2025.02.331> 

Author(s): Ducduy Nguyen, Renston Jake Fernandes, James WG Turner, David Robert Emberson

Evaluating Regional Sustainability of E-Methanol Production through GLAM-Based Life Cycle Impact Assessment, Elsevier 01/01/2025

Journal: Procedia CIRP

DOI: <https://doi.org/10.1016/j.procir.2025.01.057> 

Author(s): R.J. Fernandes, D. Nguyen, M.R. Shakeel, J.W.G. Turner

Advanced Biomass Conversion: Sustainable e-Methanol Production with Enhanced CO2 Utilization, SAE International 09/04/2024

Journal: SAE Technical Paper

DOI: <https://doi.org/10.4271/2024-01-2449> 

Author(s): Renston Jake Fernandes, Ducduy Nguyen, Mohammad Raghieb Shakeel, James W.G. Turner

Variable compression ratio hydrogen-fueled homogeneous charge compression ignition engine, SAE International 28/08/2023

Journal: SAE Technical Paper

DOI: <https://doi.org/10.4271/2023-24-0067> 

Author(s): Duc Duy Nguyen, Renston Fernandes, James WG Turner

Investigating the effect of intake temperature on hydrogen HCCI engine combustion and performance characteristics, Taylor & Francis 04/12/2023

Book: Powertrain Systems for a Sustainable Future

DOI: <http://dx.doi.org/10.1201/9781032687568-23> 

Author(s): Ducduy Nguyen, Renston Fernandes, James W.G. Turner

CONFERENCES & PRESENTATIONS - 2025

12th International Conference on Life Cycle Management 09/2025 | Palermo, Italy
Poster Presenter - "Life Cycle Assessment of E-Methanol Production from Dates"

Powertrains, Energy and Lubricants (PEL) Conference, 06/2025 | Busan, South Korea
SAE International (KSAE)
Oral Presenter - "Techno-Economic Analysis of Integrated E-Methanol Production Systems Using First- and Second-Generation Biomass"

33rd EUBCE 2025 – European Biomass Conference & Exhibition 06/2025 | Valencia, Spain
Oral Presenter - "Circular Carbon Biofuel and E-Fuel Production from Saudi Arabian Date Palm Biomass: A Process Simulation-Based Study"

13th International Conference "Fuel Science: From Production to Propulsion", RWTH Aachen University 06/2025 | Aachen, Germany
Oral Presenter - "Sustainable E-Methanol Production using Corn and its Waste in Renewable Energy Systems".

32nd CIRP Conference on Life Cycle Engineering, 04/2025 | Manchester, United Kingdom
University of Manchester
Oral Presenter - "Evaluating Regional Sustainability of E-Methanol Production through GLAM-Based Life Cycle Impact Assessment".

CORE PHD LEVEL COURSES

ME 305B - Advanced Computational Fluid Dynamics, 08/2022 – 12/2022
King Abdullah University of Science and Technology

ME 377 - Advanced Internal Combustion Engines, 01/2022 – 05/2022
King Abdullah University of Science and Technology

ENSE 344 - Modeling Bioprocess and Environmental Engineering, 08/2023 – 12/2023
King Abdullah University of Science and Technology

ACCOLADES AND SCHOLARSHIPS

KAUST Ph.D Fellowship, 10/2021
King Abdullah University of Science & Technology (KAUST)
Thuwal, Saudi Arabia

Awarded a fully funded Ph.D position at the Clean Energy Research Platform (CERP), King Abdullah University of Science & Technology (KAUST).

Sussex India Scholarship, University of Sussex 08/2018
Brighton, United Kingdom
Competitive merit-based scholarship awarded for outstanding academic performance during M.Sc. at the University of Sussex.

Research Grant (2016-2018), 06/2016
Karnataka State Council for Science & Technology (KSCST)
Mangalore, India

Graduate Student Research Grant (2017-2018)

- Secured state funding for the project "Experimental Investigation of Performance Parameters Using HFO Blended with Cooking Oil", contributing to alternative fuel research and sustainability.

Graduate Student Research Grant (2016-2017)

- Awarded state funding for the project "Conversion of Waste Plastic into Fuel Using Heliostat", aligning with circular economy principles and waste-to-energy conversion.

LANGUAGES & COMMUNICATION SKILLS

English – Very good command

Hindi – Highly proficient

Konkani – Highly proficient